

Database Systems and SQL

1. Relational Database Fundamentals

A relational database organises data into tables with rows and columns. Key concepts: Primary Key (uniquely identifies rows), Foreign Key (references another table), Index (speeds up retrieval), Constraint (enforces rules: UNIQUE, NOT NULL, CHECK).

2. ACID Properties

ACID ensures reliable transactions: Atomicity (all-or-nothing), Consistency (valid state transitions), Isolation (concurrent transactions behave serially), Durability (committed data persists after failures). Essential in finance and healthcare.

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3. PostgreSQL and pgvector

PostgreSQL is an open-source ACID-compliant relational database with 35+ years of development. pgvector adds vector similarity search: stores float vectors, supports exact and approximate nearest-neighbour search via HNSW and IVFFlat indexes. Distance operators: L2, inner product, cosine. Used for RAG systems, semantic search, and recommendations.

4. SQL Query Optimisation

Techniques: Index Usage (WHERE clauses use indexed columns), EXPLAIN ANALYZE (inspect plans), avoid SELECT * (retrieve only needed columns), JOIN optimisation, Partitioning, Materialised Views. PostgreSQL uses pg_stats for planning; run ANALYZE to update statistics.

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5. Requirements for a Production Database

Requirements: (1) High Availability - replication and failover, (2) Backup and Recovery - tested restore procedures, (3) Security - role-based access control, encryption at rest and in transit, (4) Performance - indexing and connection pooling with PgBouncer, (5) Monitoring - slow queries and lock waits, (6) Schema Migrations - versioned migrations with Alembic or Flyway, (7) Connection Limits - configure max_connections, (8) Capacity Planning - monitor growth and plan scaling.